REFERRING MULTI-OBJECT TRACKING IN MOVING-CAMERA VIDEOS VIA GLOBAL MOTION COMPENSATION

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ABSTRACT

Referring multi-object tracking (RMOT) takes a video and a natural-language expression as input and tracks all objects described by the expression. Many expressions describe how an object moves rather than how it appears. In a video captured by moving cameras, a parked vehicle may appear to move, while a vehicle moving with the camera may show little displacement. Existing motion-language RMOT methods align motion with text but do not explicitly remove camera-induced motion. In this paper, we propose extracting residual motion across frames by estimating camera motion from road-region background correspondences in driver-view videos and comparing motion features with the query language expression. We consider the motion-matching extent and integrate it with the RMOT method's prediction result through late fusion with host-specific weights. In the evaluation, we verify the performance gain by taking the proposed motion compensation module as a plug-in across different RMOT hosts. On iKUN, the module raises moving-class HOTA by 7.08 points, and all three evaluated settings exceed their hosts' published pooled HOTA.

***Index Terms*—** referring multi-object tracking, ego-motion compensation, motion-language alignment, plug-in module, Refer-KITTI, video grounding

1. INTRODUCTION

Referring multi-object tracking (RMOT) outputs tracking results of the objects described by a free-form language expression. For example, a "left-vehicles-in-red" description targets the red vehicles on the left. However, many descriptions focus on object motion, such as "moving cars" or "a vehicle turning left". In videos captured by a moving camera, the observed motion is a mixture of object motion and camera motion, and thus a static car may appear to move, while a moving car may appear static across video frames.

Previous RMOT methods [1, 2, 3] match language expressions to objects. VMRMOT [4] considers explicit motion cues, such as changes in direction and speed, in the matching process. All of these methods, however, compute motion from the object's bounding-box displacement across frames, which mixes object motion with camera motion in a video captured by moving cameras.

Conventional multi-object tracking (MOT) methods already proposed to remove camera motion: BoT-SORT [5] estimates camera-induced image motion, while UCMCTrack [6] stabilizes object trajectories. How motion compensation can be extended to RMOT is still underexplored.

We propose separating camera motion from object motion and providing motion information for language-guided tracking. It consists of four stages. First, a homography, which describes the geometric transformation across frames, is estimated to measure camera motion. The estimated homographies are accumulated across frames to reliably measure object motion. Second, residual velocity, which represents object motion after removing camera motion, is computed across frames and used as a temporal motion feature. Third, a Contrastive Motion Aligner is developed to compare motion features with language expressions and produce a motion-matching score. Finally, the motion-matching score is scaled by a host-specific weight and added to the host's matching score. Here, the host model refers to a two-stage RMOT method we use as a baseline, such as iKUN [2] and FlexHook [3]. The proposed process is taken as a plug-in to provide an extra motion-matching score. We thus improve existing RMOT baselines without retraining them.

The accumulated homographies are the key component. By compensating for camera motion, they allow the motion feature to better represent the object's actual movement rather than the apparent motion induced by camera movement. They also preserve motion information across frames, enabling the model to capture motion patterns described by expressions such as "moving" or "turning".

This work connects ego-motion compensation with motion-language modeling for RMOT. Our main contributions are as follows.

- We propose a plug-in that aligns ego-compensated residual velocity with the referring expression, thereby improving the host RMOT method without modifying or retraining it.
- Across iKUN and FlexHook, the GMC Module improves pooled HOTA in all three evaluated host settings. On iKUN, it raises MOVING HOTA from 25.531 to 32.606 ± 0.654 (approximately $+7.08$ points).

2. RELATED WORKS

2.1. Referring MOT and motion-language fusion

TransRMOT [1] introduced the RMOT task on the Refer-KITTI dataset. Later methods improved different parts of this task. iKUN [2] separated language-based object matching from object tracking, while FlexHook [3] improved how visual and language information interact. ReferGPT [7] used a multimodal large language model to describe existing object tracks, and MEX [8] provided a memory-efficient module for matching tracks with language.

More recent RMOT methods [9] further improve models' understanding of object descriptions. STORM [10] unifies grounding and tracking in an end-to-end multimodal large language model. VM-RMOT [4] explicitly represents motion using information such as position, direction, changes in distance, and changes in speed. However, these methods still estimate object motion from the object's bounding-box movements across frames. When the camera moves, the observed bounding-box movement contains both object motion and camera motion.

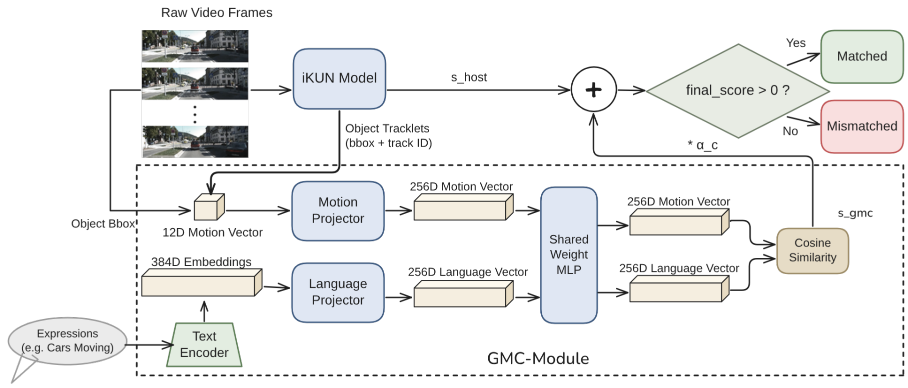

Fig. 1. The host model, iKUN in this case, provides tracking scores. The GMC Module estimates camera motion, derives residual motion, aligns it with language, and provides an extra cue to adjust the tracking scores.

2.2. Camera and ego-motion compensation in tracking

Camera motion compensation is widely used in traditional multi-object tracking (MOT). BoT-SORT [5] estimates camera-induced motion across frames. UCMCTrack [6] estimates how the camera moves relative to the road surface. EMAP [11] further separates the camera's turning and movement from object motion within the Kalman prediction process.

These methods demonstrate that compensating for camera motion can improve tracking objects across frames. However, the compensated positions are mainly used to match objects between adjacent frames. They are not accumulated over time to describe an object's motion, nor are they connected to language expressions. Our method builds on established techniques for separating camera motion from object motion, including plane-plus-parallax analysis [12], homography estimation [13], motion segmentation under moving cameras [14], and ego-motion estimation in visual odometry [15]. We do not propose a new camera-motion estimation technique. Instead, we convert camera-compensated trajectories into motion features and align them with language for RMOT.

3. METHOD

Figure 1 shows the proposed RMOT framework with the global motion compensation (GMC) Module. The GMC Module is attached to a two-stage RMOT method, iKUN [2] in this case. iKUN performs object detection and tracking and then outputs appearance-language matching scores. The inputs to the GMC Module are the sequence of object bounding boxes across frames, the video frames, and the language expression. Based on these inputs, the GMC Module estimates camera motion, extracts residual object motion, aligns it with the language expression, and produces a matching score. The score is then combined with the host model's predicted score, as an embodiment of the late fusion scheme. The module consists of four stages: ego-motion compensation, multi-scale residual velocity extraction, motion-language alignment, and late fusion.

3.1. Ego-motion compensation

This stage estimates camera motion from consecutive video frames so that object motion can be separated from ego-motion. Refer-KITTI is recorded from a driver-view camera, so the road surface is visible in most frames and stays close to a single plane. A homography is exact for a plane, which makes the road a reliable reference for camera motion. We select Shi-Tomasi corners [16] in the lower half of the frame, excluding the ~~region where~~ interiors of the previous frame's tracked object boxes~~overlap~~, and follow them into the next frame with pyramidal Lucas-Kanade optical flow [17]. RANSAC [18] then estimates the frame-to-frame homography $H_{t-1\to t}$ from these correspondences. ~~Sparse road texture can make this fit fail; the module then falls back to a global homography estimated from ORB [19] features over the whole frame, with object boxes masked out~~For a forward-facing camera the horizon lies near the vertical image center: pixels above it belong to buildings and sky, which are not on the road plane, whereas the lower half contains the road from directly ahead of the vehicle to the horizon. We therefore fit the homography to the lower half only.

To describe camera motion across frames, the estimated homographies are accumulated as

$$H_{t-k\to t} = H_{t-1\to t} \times H_{t-2\to t-1} \times \cdots \times H_{t-k\to t-k+1}, \quad (1)$$

where $H_{t-k\to t}$ represents the accumulated camera transformation from frame I_{t-k} to frame I_t. Instead of repeatedly transforming already warped coordinates, the accumulated homography is applied only once to the original object centroid. This reduces numerical drift and provides a stable reference for estimating ego velocity in the next stage.

3.2. Multi-scale residual velocity

Using the accumulated homography matrix and the objects' bounding boxes, this stage estimates the residual velocity, which represents object motion after removing the estimated camera motion. For a temporal gap g, the raw velocity ~~v_g^{raw} is derived~~ $v_g^{\text{raw}} = (\boldsymbol{o}_t - \boldsymbol{o}_{t-g})/g$ is the observed image-plane velocity of an object, computed from the displacement of ~~the object centroid between two frames (~~its centroid $\boldsymbol{o}$ between frames I_{t-g} and I_t~~) and is therefore mixed with objectmotion and camera motion~~; it is the sum of the object's own motion and the motion induced by the camera. To remove the camera motion, the accumulated homography $H_{t-g\to t}$ is applied to ~~an object's~~ the centroid $\boldsymbol{o}_{t-g}$ ~~at I_{t-g}~~ in homogeneous coordinates, yielding $\hat{\boldsymbol{o}}_t$, the centroid predicted at I_t under the assumption that the object is stationary. The ego ~~displacement $\hat{\boldsymbol{o}}_t - \boldsymbol{o}_{t-g}$ thus arises from~~ velocity $v_g^{\text{ego}} = (\hat{\boldsymbol{o}}_t - \boldsymbol{o}_{t-g})/g$ is thus the image-plane velocity induced by camera motion alone, ~~and dividing it by the temporal gap g yields the ego velocity v_g^{ego}~~ evaluated at the object's location. The residual velocity v_g^{res} is ~~then computed as~~ the camera-motion-compensated object velocity, obtained by subtracting the ego velocity from the raw velocity:

$$v_g^{\text{res}} = v_g^{\text{raw}} - v_g^{\text{ego}} \cancel{,} = \big(\boldsymbol{o}_t - \hat{\boldsymbol{o}}_t\big)/g. \tag{2}$$

This representation approximates the motion that a stationary camera would observe. For example, a parked vehicle has near-zero residual velocity even when camera motion causes a large image displacement. All velocities are normalized by the image dimensions to make them independent of image resolution. In the implementation the factor $1/g$ is omitted; this is equivalent up to a per-gap rescaling of the first projection layer of Sec. 3.3.

To capture motion over different time spans, residual velocities are computed at temporal gaps of $\{2, 5, 10\}$ frames. That means the accumulated homography matrices mentioned in Eqn. (1) have three versions, i.e., $H_{t-2\to t}$, $H_{t-5\to t}$, and $H_{t-10\to t}$. The residual velocities are concatenated with the object's box information to form a 12-dimensional motion feature,

$$\big((dx_s, dy_s), (dx_m, dy_m), (dx_l, dy_l), \Delta w, \Delta h, c_x, c_y, w, h\big), \tag{3}$$

where the first six values are the residual velocities at the three temporal gaps, and $\Delta w, \Delta h$ capture the change in box scale. The terms c_x and c_y describe the coordinates of the box center, and w and h describe the width and height of the box, respectively.

A track contributes a motion feature only once it has 11 contiguous frames of history, the shortest history at which all three gaps are defined; on shorter histories the module abstains and the host score passes through unchanged. The threshold follows from the largest gap and adds no free parameter.

3.3. Motion-language alignment

Given the motion features mentioned above and the language expression, this stage measures how well the observed object motion matches the described motion. The 12-dimensional motion feature is encoded by a motion encoder, while the language expression is encoded by all-MiniLM-L6-v2 (MiniLM) [20] into the text embeddings. Both embeddings are projected into the same feature space using linear projection layers and a shared Multi-Layer Perceptron (MLP).

The model is trained using Information Noise Contrastive Estimation (InfoNCE) [21] loss to maximize the similarity between matched motion-expression pairs while separating unmatched pairs. A positive pair is formed by the embeddings of an object and the corresponding expression. Other object-expression pairs in the same mini-batch are treated as negatives. The MiniLM text encoder is kept fixed, and only the projection layers and motion encoder are trained. During inference, the cosine similarity s_{gmc} between an object's motion embeddings and the target expression's embeddings is used as the matching score and passed to the late fusion stage described below.

3.4. Fusion

The matching score is combined with the prediction score of the host RMOT model to obtain the final prediction:

$$s_{\text{final}} = s_{\text{host}} + \alpha_c\, s_{\text{gmc}}, \tag{4}$$

where s_{host} is the prediction score produced by the host model and $c \in \{\text{MOVING}, \text{STATIC}, \text{APPEARANCE}\}$ is the category of the referring expression, assigned by a keyword classifier. MOVING and STATIC expressions use the motion-oriented weight α_{mot}, while APPEARANCE expressions use the appearance-oriented weight α_{app}. Because host models produce scores over different ranges, both weights are selected per host by leave-one-sequence-out cross-validation: for each fold the weight with the highest pooled HOTA is chosen on all but one sequence, and the final value is the median over folds. This yields $(\alpha_{\text{mot}}, \alpha_{\text{app}}) = (0.7, 0.1)$ on iKUN, whereas on both FlexHook settings the procedure selects $\alpha_{\text{mot}} = \alpha_{\text{app}}$ (7 on V1 and 5 on V2), so the routing degenerates to a single weight. Setting $\alpha_c = 0$ recovers the native host exactly.

Since the GMC Module only adds extra influence on the final prediction score, it can be integrated into existing two-stage RMOT frameworks without changing or retraining their detector, tracker, visual encoder, or language encoder.

4. EXPERIMENTS

4.1. Setup

We evaluate the proposed module on the Refer-KITTI V1 [1] and the Refer-KITTI V2 [22] benchmarks using the HOTA metric from TrackEval [23], which jointly measures detection and association performance. We report both moving-class HOTA, which evaluates only expressions referring to moving objects, and pooled HOTA, which evaluates all expressions in the benchmark. The V1 test split contains three sequences and the V2 test split four.

The proposed module is attached to two host models, iKUN [2] and FlexHook [3]. For comparison, we also evaluate a raw-velocity variant that uses image-plane motion directly, without ego-motion compensation. The GMC Module is trained using AdamW [24] with a learning rate of 10^{-3}, batch size 256, and 100 epochs. The aligner is trained on the Refer-KITTI V1 training split (15 sequences, disjoint from the test sequences). Each positive pair consists of an expression and the motion feature of a ground-truth object annotated as its referent, computed from the object's ground-truth boxes; tracker outputs are used only at test time. The road-plane homography tracks up to 600 corners with a 3-pixel RANSAC threshold ~~, while the global homography uses 1,500 ORB keypoints with a 5-pixel threshold. Across the four evaluation sequences, the road-plane homography~~ and succeeds on all ~~2,065~~ 7,690 adjacent frame pairs ~~, so the global homography is never used as a fallback~~of the 19 training and evaluation sequences. Excluding host inference, the module runs at ~~31.8~~ 149 FPS on CPU (6.7 ms per frame).

Results are averaged over three random seeds for the main configurations; ablation analyses use five seeds together with Welch's t-test. Both host models use their original detection results; FlexHook V1 is evaluated under the official 150-expression protocol. Our reproduced FlexHook baselines match their published scores exactly at α_c=0, while our reproduced iKUN (44.224) is slightly below its published score; Table 1 therefore reports the stricter comparison against the published baselines.

Refer-KITTI does not provide a dedicated validation split. All fusion weights are therefore selected by the leave-one-sequence-out procedure of Sec. 3.4, so no weight is chosen on the sequence it is evaluated on.

Refer-KITTI is highly imbalanced, with only about one-sixth of the language expressions referring to moving objects. Consequently, improvements in motion understanding may have only a limited impact on pooled HOTA. Therefore, moving-class HOTA is used as the primary evaluation metric, while pooled HOTA is reported to reflect overall benchmark performance.

Table 1. Performance in terms of the pooled HOTA on the Refer-KITTI test dataset.

Host	Published	GMC Module	Gain
iKUN [2]	44.564	44.847 ± 0.107	$+0.283$
FlexHook V1 [3]	53.824	53.980 ± 0.059	$+0.156$
FlexHook V2 [3]	42.526	42.625 ± 0.032	$+0.099$

Table 2. Moving-class HOTA on movement-related expressions (27/158 for iKUN, 25/150 for FlexHook V1, 136/862 for FlexHook V2), n=3 seeds.

Host	Baseline	Performance gain
iKUN	25.53	$+7.08 \pm 0.65$
FlexHook V1	44.31	$+0.67 \pm 0.13$
FlexHook V2	38.15	$+0.18 \pm 0.06$

4.2. Main results

Table 1 summarizes the pooled HOTA results on the Refer-KITTI dataset. We evaluate the performance variations when iKUN or FlexHook is used as the host model, with or without the GMC Module. With the module attached, all three settings exceed not only our reproduced natives but also the hosts' published pooled HOTA ($+0.283$, $+0.156$, and $+0.099$, respectively).

Table 2 reports HOTA variations for movement-related expressions. All three host settings improve, and the size of the improvement differs by an order of magnitude: iKUN gains $+7.08 \pm 0.65$, FlexHook V1 $+0.67 \pm 0.13$, and FlexHook V2 $+0.18 \pm 0.06$.

4.3. Ablation Studies

Table 3 reports the five-seed ablation measurements on iKUN. The full module improves moving-class HOTA from 25.53 to 32.30 ± 0.63 and pooled HOTA from 44.22 to 44.80 ± 0.10. The full module also improves STATIC HOTA from 43.91 to 44.55; without ego compensation it falls to 43.70, below the native host, while removing the multi-scale gaps gives 44.43. Removing ego-motion compensation costs -3.81 moving-class and -0.64 pooled HOTA; removing the multi-scale gaps costs -1.85 and -0.32. Both drops are significant under Welch's t-test ~~, and~~ (n=5; ego: t=11.8 moving-class, 11.7 pooled; multi-scale: t=6.2 and 6.7; all p<0.01), and ego-motion compensation accounts for the larger share.

Figure 2 illustrates this behavior using a real example. The parked car moves across the image because of camera motion, while the moving car remains at nearly the same image location. Without ego-motion compensation, these two cases are easily confused because they have similar image-plane motion. After compensating for camera motion, the parked car has nearly zero residual motion, while the moving car retains nonzero residual motion. This allows the GMC Module to correctly distinguish between stationary and moving objects.

Table 3. Per-class HOTA ablation on iKUN (n=5 seeds).

Variant	MOVING HOTA	POOLED HOTA
Native host (no module)	25.53	44.22
Full module	**32.30 ± 0.63**	**44.80 ± 0.10**
−ego (raw velocity)	28.49 ± 0.35	44.17 ± 0.07
−multiscale (single gap)	30.45 ± 0.20	44.49 ± 0.02

Fig. 2. "Moving cars" (Refer-KITTI seq 0005). The parked car (red) sweeps across the image as the camera passes, while the moving car (green) barely shifts. Raw image motion misclassifies both; the ego-compensated residual recovers both from the same host score.

5. CONCLUSION AND LIMITATIONS

The GMC Module is a decision-level plug-in that improves motion grounding in referring multi-object tracking without modifying the detector, tracker, or appearance-language matching model. By compensating for camera motion and aligning residual motion with language descriptions, the proposed module helps distinguish object motion from camera-induced motion. Road-region background correspondences provide a geometric reference for estimating camera motion in driver-view driving scenes, after which the residual motion is aligned with language and integrated through late fusion with host-specific weights. On Refer-KITTI, pooled HOTA improves in all three evaluated host settings. For movement-related expressions, the largest gain is observed on iKUN ($+7.08 \pm 0.65$), and the ablation results show improvements in both MOVING and STATIC HOTA for the full module.

Despite these promising results, several limitations remain. First, the road-plane homography assumes a flat ground ~~, and falls back to the global estimate, which is less accurate, when it isn't~~plane; scenes with strong slopes or uneven terrain fall outside the model. Second, a keyword classifier decides which weight an expression receives. Therefore, a misread expression gets the incorrect fusion weight: 14 of the 126 direction expressions in Refer-KITTI V1 are treated as appearance. Third, two weights do not help every host. On both FlexHook settings the leave-one-sequence-out procedure selects a single weight. Finally, our experiments are conducted on two RMOT frameworks and the Refer-KITTI benchmarks, and additional evaluations on more host architectures and datasets are needed to further validate the generality of the proposed approach.

In future work, we plan to investigate more robust camera motion estimation methods and richer motion features that can better handle complex dynamic scenes. We also hope to evaluate the GMC Module on a broader range of referring multi-object tracking frameworks to further improve its generalization.

6. REFERENCES

[1] Dongming Wu, Wencheng Han, Tiancai Wang, Xingping Dong, Xiangyu Zhang, and Jianbing Shen, "Referring multi-object tracking," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 2023, pp. 14633–14642, IEEE.

[2] Yunhao Du, Cheng Lei, Zhicheng Zhao, and Fei Su, "iKUN: Speak to trackers without retraining," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 2024, pp. 19135–19144, IEEE.

[3] Weize Li, Yunhao Du, Qixiang Yin, Zhicheng Zhao, and Fei Su, "Rethinking two-stage referring-by-tracking in referring multi-object tracking: Make it strong again," 2025, Accepted to CVPR 2026.

[4] Weiyi Lv, Ning Zhang, Hanyang Sun, Haoran Jiang, Kai Zhao, Jing Xiao, and Dan Zeng, "Vision–motion–reference alignment for referring multi-object tracking via multi-modal large language models," 2025.

[5] Nir Aharon, Roy Orfaig, and Ben-Zion Bobrovsky, "BoT-SORT: Robust associations multi-pedestrian tracking," 2022.

[6] Kefu Yi, Kai Luo, Xiaolei Luo, Jiangui Huang, Hao Wu, Rongdong Hu, and Wei Hao, "UCMCTrack: Multi-object tracking with uniform camera motion compensation," in *Proceedings of the AAAI Conference on Artificial Intelligence*. 2024, vol. 38, pp. 6702–6710, AAAI Press.

[7] Tzoulio Chamiti, Leandro Di Bella, Adrian Munteanu, and Nikos Deligiannis, "Refergpt: Towards zero-shot referring multi-object tracking," 2025, CVPR Workshops.

[8] Huu-Thien Tran, Phuoc-Sang Pham, Thai-Son Tran, and Khoa Luu, "Mex: Memory-efficient approach to referring multi-object tracking," 2025.

[9] Wenyan He, Yajun Jian, Yang Lu, and Hanzi Wang, "Visual-linguistic representation learning with deep cross-modality fusion for referring multi-object tracking," in *ICASSP 2024 – 2024 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2024, pp. 6310–6314.

[10] Zijia Lu, Jingru Yi, Jue Wang, Yuxiao Chen, Junwen Chen, Xinyu Li, and Davide Modolo, "STORM: End-to-end referring multi-object tracking in videos," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), Findings*, 2026.

[11] Navid Mahdian, Mohammad Jani, Amir M. Soufi Enayati, and Homayoun Najjaran, "Ego-motion aware target prediction module for robust multi-object tracking," 2024.

[12] Michal Irani and P. Anandan, "A unified approach to moving object detection in 2D and 3D scenes," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 20, no. 6, pp. 577–589, 1998.

[13] Richard Hartley and Andrew Zisserman, *Multiple View Geometry in Computer Vision*, Cambridge University Press, 2nd edition, 2004.

[14] Pia Bideau and Erik Learned-Miller, "It's moving! a probabilistic model for causal motion segmentation in moving camera videos," in *Computer Vision – ECCV 2016*. 2016, vol. 9912 of *Lecture Notes in Computer Science*, pp. 433–449, Springer.

[15] Davide Scaramuzza and Friedrich Fraundorfer, "Visual odometry [tutorial]," *IEEE Robotics & Automation Magazine*, vol. 18, no. 4, pp. 80–92, 2011.

[16] Jianbo Shi and Carlo Tomasi, "Good features to track," in *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 1994, pp. 593–600.

[17] Bruce D. Lucas and Takeo Kanade, "An iterative image registration technique with an application to stereo vision," in *International Joint Conference on Artificial Intelligence (IJCAI)*, 1981, pp. 674–679.

[18] Martin A. Fischler and Robert C. Bolles, "Random sample consensus: A paradigm for model fitting with applications to image analysis and automated cartography," *Communications of the ACM*, vol. 24, no. 6, pp. 381–395, 1981.

[19] Ethan Rublee, Vincent Rabaud, Kurt Konolige, and Gary Bradski, "ORB: An efficient alternative to SIFT or SURF," in *2011 International Conference on Computer Vision (ICCV)*. 2011, pp. 2564–2571, IEEE.

[20] Nils Reimers and Iryna Gurevych, "Sentence-BERT: Sentence embeddings using siamese BERT-networks," in *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, Hong Kong, China, 2019, pp. 3982–3992, Association for Computational Linguistics.

[21] Aaron van den Oord, Yazhe Li, and Oriol Vinyals, "Representation learning with contrastive predictive coding," 2018.

[22] Yani Zhang, Dongming Wu, Wencheng Han, and Xingping Dong, "Bootstrapping referring multi-object tracking," 2024, Introduces TempRMOT and Refer-KITTI-V2.

[23] Jonathon Luiten, Aljoša Ošep, Patrick Dendorfer, Philip Torr, Andreas Geiger, Laura Leal-Taixé, and Bastian Leibe, "HOTA: A higher order metric for evaluating multi-object tracking," *International Journal of Computer Vision*, vol. 129, no. 2, pp. 548–578, 2021.

[24] Ilya Loshchilov and Frank Hutter, "Decoupled weight decay regularization," in *International Conference on Learning Representations (ICLR)*, 2019.